

PHASE 10: TESTING AND DEBUGGING FINAL REPORT

Project Name: Academic Credential Verification & Alumni Tracking System

Student Name: Dusabe Marie Rose

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Document Status: Final Version for Submission 1. Executive Summary

This report presents the final activities completed during Phase 10: Testing and Debugging from 11 April to 17 April 2026. The purpose of this phase was to test the complete system, identify technical problems, improve performance, validate security, and ensure that both frontend and backend components function correctly before final submission. At the end of this phase, the system was confirmed to be stable, secure, responsive, and ready for final presentation.

2. Testing Schedule and Activities

Day 1: Test Planning and Strategy Preparation – Completed

Day 2: Backend Unit Testing – Completed

Day 3: API End-to-End Testing with Postman – Completed

Day 4: Security and Role-Based Access Testing – Completed

Day 5: Bug Fixing and Performance Improvement – Completed

Day 6: Regression Testing and Re-validation – Completed

Day 7: Final Evidence Collection and Documentation – Completed

3.1 Test Planning and Strategy Preparation

Success criteria were defined: successful login, blocked unauthorized access, accurate database storage, smooth dashboard loading, correct verification results, and good response time.

3.2 Backend Unit Testing

Main services including AlumniService, UserService, CredentialService, and repositories were tested. CRUD operations worked correctly and data integrity was maintained.

3.3 API End-to-End Testing (Postman)

REST endpoints were tested using Postman:

POST /api/auth/register – user registration successful.

POST /api/auth/login – token returned successfully.

GET /api/alumni/profile – protected endpoint accessible with token.

POST /api/credentials/verify – verification process successful.

3.4 Authentication and Security Testing

JWT token validation was tested. Protected routes denied unauthorized users. Role-based access control worked for ADMIN, ALUMNI, and EMPLOYER roles. Password encryption using BCrypt was verified.

3.5 Frontend Testing

React pages including Login, Register, Dashboard, Profile, Alumni, and Verification pages were tested. Navigation using React Router worked correctly. Error messages and success notifications displayed properly.

3.6 Bug Fixing and Optimization

Issue: Port already in use – fixed by terminating background Java process.

Issue: Empty dashboard charts – fixed with null checks and fallback values.

Issue: Image upload size error – multipart limit increased to 10MB.

Issue: Slow API calls – optimized queries and removed unnecessary reloads.

3.7 Regression Testing

After fixes, the entire system was tested again to ensure no previous features were broken. All major functions continued to work correctly.

4. Deliverables Checklist

- Full Spring Boot Backend verified
- React Frontend verified
- Postman Collection completed
- Documentation completed
- Evidence screenshots collected

5. Conclusion

The testing phase was successful. The Academic Credential Verification & Alumni Tracking System is resilient, secure, user-friendly, and ready for final presentation and deployment. Finalized by: Dusabe Marie Rose

Date: 17 April 2026